清华大学本科生考试试题专用纸

考试课程：统计推断　考试时间：2022 年 12 月 22 日 19:20-21:45 页数：2 页

$\chi^2_{1,0.05} = 3.84$, $\chi^2_{2,0.05} = 5.99$, $\chi^2_{3,0.975} = 0.22$, $\chi^2_{3,0.025} = 9.35$, $\chi^2_{3,0.05} = 7.81$, $\chi^2_{4,0.05} = 9.49$, $\chi^2_{5,0.05} = 11.07$, $\chi^2_{6,0.05} = 12.59$, $\chi^2_{200,0.05} = 234$, $\chi^2_{200,0.025} = 241$, $\chi^2_{200,0.95} = 168$, $\chi^2_{200,0.975} = 163$, $t_{4,0.025} = 2.78$, $t_{4,0.01266} = 3.48$, $t_{4,0.0125} = 3.50$, $t_{5,0.025} = 2.57$, $t_{5,0.01266} = 3.15$, $t_{5,0.0125} = 3.16$, $t_{7,0.025} = 2.36$, $t_{7,0.05} = 1.89$, $t_{9,0.025} = 2.26$, $t_{9,0.05} = 1.83$, $t_{15,0.025} = 2.13$, $t_{15,0.05} = 1.75$, $t_{16,0.025} = 2.12$, $t_{23,0.05} = 1.71$, $z_{0.05} = 1.64$, $z_{0.025} = 1.96$, $F_{4,5,0.025} = 7.39$, $F_{4,5,0.975} = 0.11$, $F_{5,6,0.025} = 5.99$, $F_{5,6,0.975} = 0.14$.

1 (30') The independent random samples $X_i,\ i = 1, \ldots, 6$ and $Y_j,\ j = 1, \ldots, 5$ represent resistance measurements taken on two test pieces, and the observed values (in ohms) are as follows:

$$x_1 = 35,\ x_2 = 40,\ x_3 = 41,\ x_4 = 37,\ x_5 = 38,\ x_6 = 41,$$

$$y_1 = 38,\ y_2 = 39,\ y_3 = 41,\ y_4 = 42,\ y_5 = 38.$$

Suppose that X_i i.i.d. $\sim N(\mu_1, \sigma_1^2)$ and Y_j i.i.d. $\sim N(\mu_2, \sigma_2^2)$, where $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2$ are unknown, $\sigma_1^2 > 0$, and $\sigma_2^2 > 0$.

(i) (5') Construct a 95% confidence interval for μ_2;

(ii) (5') Construct a 95% confidence region for (μ_1, μ_2);

(iii) (5') Testing the hypothesis $H_0 : \mu_1 = 39 \longleftrightarrow H_1 : \mu_1 \neq 39$ at level of significance $\alpha = 0.05$;

(iv) (5') Testing the hypothesis $H_0 : \sigma_1^2 = \sigma_2^2 \longleftrightarrow H_1 : \sigma_1^2 \neq \sigma_2^2$ at level of significance $\alpha = 0.05$;

(v) (5') Construct a 95% confidence interval for $\mu_1 - \mu_2$;

(vi) (5') Testing the hypothesis $H_0 : \mu_1 \leq \mu_2 \longleftrightarrow H_1 : \mu_1 > \mu_2$ at level of significance $\alpha = 0.05$.

2 (10') Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from a distribution with p.d.f $f(x; \theta) = e^{-(x-\theta)}$, $x > \theta$, where $-\infty < \theta < +\infty$ is unknown.

(i) (5') Derive the distribution of $X_{(1)} - \theta$ and show that it does not depend on θ;

(ii) (5') Construct an $1 - \alpha$ upper confidence limit of θ.

3 (10') Let $X_1, \ldots, X_m$ be a random sample from normal distribution $N(\mu_1, \sigma^2)$ and $Y_1, \ldots, Y_n$ be a random sample from $N(\mu_2, \sigma^2)$, where $\mu_1, \mu_2 \in R$ and $\sigma^2 > 0$ are unknown. Suppose that X_i's and Y_j's are independent. Testing the following hypothesis at level of significance α:

$$H_0 : \mu_1 = 2\mu_2 \longleftrightarrow H_1 : \mu_1 \neq 2\mu_2.$$

4. (15') Two testers (A and B) analyzed the same product samples and obtained the results in Table 1. Test whether there is a significant difference (in means) of the analysis results of testers A and B. Take $\alpha = 0.05$.

Table 1: Testing results

tester	sample 1	sample 2	sample 3	sample 4	sample 5	sample 6	sample 7	sample 8
A	4.1	3.5	3.4	3.6	3.8	4.5	3.4	3.7
B	3.5	4.0	3.6	3.7	4.2	3.8	2.9	4.1

5. (10') Let $\boldsymbol{X}_i = (X_{i1}, \cdots, X_{in_i})$ be a random sample of size n_i from Normal distribution $N(\mu_i, \sigma^2)$, where $\mu_i \in R$ and $\sigma^2 > 0$ are unknown, $i = 1, \cdots, m$, and suppose that $\boldsymbol{X}_i$'s are independent. Derive the likelihood ratio test at level of significance α:

$$H_0 : \mu_1 = \cdots = \mu_m \longleftrightarrow H_1 : \mu_1, \ldots, \mu_m \text{ are not all equal.}$$

6. (25') Let $\boldsymbol{X} = (X_1, \cdots, X_n)$ be a random sample from exponential distribution with p.d.f. $f(x;\theta) = \theta e^{-\theta x},\ x > 0,\ \theta \in \Theta = (0, +\infty)$.

   (i) (5') Show that $f(\cdot;\theta)$ is an exponential family;

   (ii) (5') Show that $Y = 2\theta \sum_{i=1}^{n} X_i$ is distributed as χ^2_{2n};

   (iii) (10') Derive the UMP test for testing the hypothesis $H_0 : \theta \leq \theta_0 \ \longleftrightarrow\ H_1 : \theta > \theta_0$ at level of significance α ($\theta_0 > 0$);

   (iv) (5') Carry out the UMP test in (iii) for $\theta_0 = 10$, $\alpha = 0.05$, $n = 100$, and give the formula of the power of the test at $\theta = 15$.

   – Note: The p.d.f. of χ^2_r is $1/\{\Gamma(r/2)2^{r/2}\}x^{r/2-1}e^{-x/2}$, $x > 0$. The characteristic functions of χ^2_r and exponential distribution are $(1-2it)^{-r/2}$ and $\theta/(\theta - it)$, respectively, where $i^2 = -1$.